

ESSEN MULHEIM, Germany

WMO: 104100

Lat: 51.4042N

Lon: 6.9678E

Elev: 150

StdP: 99.54

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-9.9	-6.9	-13.7	1.2	-7.7	-11.4	1.4	-5.4	12.2	6.7	11.0	4.1	4.3	60	0.544

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	28.2	19.3	26.6	18.4	24.8	17.6	20.1	26.2	19.2	25.1	18.4	23.6	3.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	13.2	22.4	17.1	12.4	21.2	16.3	11.8	20.7	58.3	26.4	55.1	24.7	52.4	23.7	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	8.4	7.4	-9.8	31.4	5.4	1.7	-13.7	32.6	-16.8	33.6	-19.9	34.5	-23.8	35.7		
(5)			-10.3	21.9	5.2	0.8	-14.1	22.5	-17.1	22.9	-20.1	23.3	-23.8	23.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	2.1	2.0	5.5	8.4	13.4	15.4	18.0	17.8	14.6	10.9	6.2	4.1		
	DBStd	6.78	4.79	4.74	3.58	3.75	3.94	3.70	3.49	3.13	3.11	3.30	3.70	3.48		
	HDD10.0	1038	244	224	147	74	12	2	0	0	1	29	122	183		
	HDD18.3	3178	502	457	399	298	161	105	51	49	120	231	363	440		
	CDD10.0	1005	0	1	6	26	116	163	248	241	138	57	8	2		
	CDD18.3	103	0	0	0	0	7	16	41	32	7	1	0	0		
	CDH23.3	716	0	0	0	1	51	103	310	218	33	1	0	0		
	CDH26.7	144	0	0	0	0	4	13	75	49	3	0	0	0		
(14) Wind	WSAvg	3.6	4.6	4.3	4.2	3.7	3.2	3.1	3.0	2.9	3.1	3.4	3.8	4.1		
(15) Precipitation	PrecAvg	920	79	59	67	61	73	88	89	87	71	72	84	93		
	PrecMax	1220	156	162	157	126	167	191	189	196	172	164	147	212		
	PrecMin	645	4	5	12	8	13	12	15	28	26	19	6	16		
	PrecStd	140	38	34	37	30	33	41	44	42	33	36	34	44		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	14.5	18.3	21.9	27.0	28.3	31.3	30.6	26.9	21.8	15.7	13.2		
		MCWB	9.8	10.3	10.8	13.6	16.9	19.7	20.2	19.4	18.4	16.3	10.9	10.5		
	2%	DB	10.0	11.8	14.7	18.8	24.8	26.2	28.7	27.9	23.8	19.0	13.9	11.9		
		MCWB	8.5	9.0	9.7	11.9	15.5	18.4	19.9	19.0	17.5	14.4	11.0	9.8		
	5%	DB	8.8	9.7	11.9	16.5	22.5	24.2	26.8	25.7	21.5	17.1	12.5	10.2		
		MCWB	7.5	7.5	8.4	10.6	14.7	17.2	18.7	18.2	16.5	13.6	10.2	8.9		
	10%	DB	7.8	8.0	10.2	14.7	20.4	22.1	24.7	23.7	19.5	15.3	11.3	8.8		
		MCWB	6.5	6.3	7.6	9.8	13.8	16.0	18.1	17.3	15.7	12.6	9.7	7.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	10.9	11.5	14.2	17.9	21.2	21.5	20.6	19.7	17.1	12.7	11.2		
		MCDB	11.1	13.3	16.7	20.1	25.7	26.3	28.3	27.2	24.7	21.0	14.1	12.4		
	2%	WB	8.9	9.4	10.3	12.5	16.3	19.3	20.5	19.4	18.2	15.3	11.7	9.9		
		MCDB	9.6	11.2	13.5	16.9	22.7	24.7	26.8	26.0	22.1	18.0	12.8	11.1		
	5%	WB	7.7	7.8	9.1	11.4	15.2	18.0	19.5	18.7	17.2	14.1	10.7	9.0		
		MCDB	8.5	9.4	11.4	15.6	21.3	23.1	25.4	24.5	20.5	16.3	12.0	10.3		
	10%	WB	6.5	6.5	8.0	10.1	14.2	16.8	18.5	17.9	16.2	13.1	9.8	7.9		
		MCDB	7.7	7.9	9.9	14.1	19.3	20.9	23.9	22.5	19.0	14.9	11.3	8.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.2	5.1	5.8	7.2	8.1	7.6	8.2	8.2	6.8	5.7	4.6	3.7		
		MCDBR	5.1	7.0	8.0	9.7	11.7	11.1	11.4	11.5	9.6	7.9	5.9	5.0		
	5% WB	MCWBR	4.4	5.2	5.0	5.2	5.7	5.4	5.3	5.1	4.7	4.3	4.3	4.1		
		MCWBR	4.8	6.5	6.9	8.8	10.5	9.9	10.3	10.2	8.4	6.5	5.3	4.8		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.344	0.373	0.408	0.413	0.413	0.421	0.414	0.386	0.359	0.338	0.317		
		taud	2.412	2.374	2.306	2.238	2.259	2.276	2.253	2.299	2.366	2.434	2.416	2.385		
	Ebn at Noon		694	775	817	828	842	844	828	814	796	748	663	630		
		Edh at Noon	61	81	104	125	128	127	128	117	98	76	60	54		
(44) All-Sky Solar Radiation	RadAvg		0.73	1.40	2.52	4.06	4.91	5.29	5.03	4.28	3.13	1.84	0.87	0.56		
	RadStd		0.11	0.27	0.38	0.46	0.45	0.47	0.63	0.40	0.37	0.17	0.11	0.07		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (78 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page